METRIC

MIL-S-322C
INTERIM AMENDMENT 1 (OS)
30 April 1990

MILITARY SPECIFICATION

SODIUM NITRATE

This interim amendment is approved for use within the Naval Sea Systems Command(OS), with MIL-S-322C, dated 5 February 1968.

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Add the following new paragraphs with new table III:

"4.4.5.4 Alternate method - ferrous sulfate titration.

4.4.5.4.1 Apparatus. The following apparatus or equivalent shall be used:

a. Fisher Titrimeter II System with controller, pH meter, buret-dispenser with 25-ml amber syringe, stirrer, recorder, and platinum and glass electrodes,

b. Nitrogen purge on titrant reservoir, 2-ounce-per-square-inch,

c. Circulating-cooling bath and a small external bath with coolant maintained between 2° to 5°C.

4.4.5.4.2 Chemicals. The following chemicals shall be used:

a. Sulfuric acid (H_2SO_4), 95-96 %,

b. Potassium nitrate (KNO_3),* NIST, 99.6%,

c. Ferrous sulfate ($FeSO_4$), 0.4 % nitrogen per ml. Dissolve 350 gm of $FeSO_4.7H_2O$ in 800 ml distilled water. Slowly add one liter cold 1:1 H_2SO_4 and dilute to 2 liters with water. Preserve titrant with nitrogen purge. Allow solution to stand for 3 days.

*NIST= National Institute of Standards and Technology

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4.4.5.4.3 Standardization of the ferrous sulfate.

Note: If the system has been down 8 hours, purge the dispensing system. Condition the electrodes in H_2SO_4 for 4 hours. Zero the pH meter by setting the range to 1000 millivolts (mV) according to the manufacturer's instructions.

Grind the KNO_3 standard to the consistency of table salt. Pre-weigh 0.50 ± 0.05 gm KNO_3 in a weighing bottle. Dry in oven at 135°C for 4 hours. If the KNO_3 has been out of the oven for more than 4 hours, redry at 105°C for 2 hours. Store in a desiccator.

Set controller functions (see table III) for a dead-stop potentiometric titration. In a hot-dry beaker, add a stir bar and 150 ml H_2SO_4. Place beaker in bath on stirrer. Ensure coolant is circulating through the external bath. Weigh KNO_3 to the nearest 0.1 mg and slowly pour into vortex of the acid. Lower the electrodes and delivery tip into acid. Turn meter to 1000 mV. When sample has dissolved, begin titration. Adjust titrant addition rate to obtain an endpoint within 8 to 10 minutes. After endpoint is reached and held for one minute, turn meter to stand-by. Rinse electrodes in H_2SO_4. Calculate the nitrogen equivalence factor (F):

$$\text{Factor} = \frac{13.855 \times W}{V}$$

Where:

F = factor, % N/ml
W = weight of KNO_3, gm
V = volume of titrant, ml.

Perform standardization in duplicate. Checks should agree within 0.0005.

TABLE III. Recommended instrument settings.

METER	temp - 25°C function - stand-by or 1000 mV
ENDPOINT	350 mV
PROP BAND	85
MIN DEL	0

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4.4.5.4.4 Procedure. Grind a portion of sample to the consistency of table salt. Weigh 0.50 ± 0.05 gm of sample in a weighing bottle. Dry at 150°C for 2 hours. Repeat 4.4.5.4.3 with the $NaNO_3$. Calculate the nitrogen content of the $NaNO_3$:

$$\text{Percent sodium nitrate} = \frac{6.07\ V \times F}{W}$$

Where:

V = volume of titrant, ml
F = equivalence factor, %N/ml
W = weight of $NaNO_3$, gm.

Perform test in triplicate. Checks should agree within 0.03 percent. Record average value to the nearest 0.01 percent."

Preparing Activity:
Navy-OS
(Project 6810-N062)

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